Lab 8 Assignment: Numpy

Instructions

Answer the following questions using NumPy functions. You are only given the inputs and expected outputs. Implement your solutions and verify if the outputs match.

Questions

1. **Raise Elements of an Array to a Power without using iterations:** Given a NumPy array, raise each element to a specified power. Do not use iterations like `for`, `while` etc.

   Input:

   ```
   arr = np.array([1, 2, 3, 4, 5])
   power = 3
   ```

   Expected Output:

   ```
   [  1   8  27  64 125]
   ```

2. **Interchanging Columns of a 2D Array:** Given a 2D array, interchange the i-th and j-th columns. Obtain i and j as inputs from user and print the result.

   Input:

   ```
   arr = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [7, 8, 9]])
   i, j = 0, 2
   ```

   Expected Output:

   ```
   [[3 2 1]
    [6 5 4]
    [9 8 7]]
   ```

3. **Find Indices Where Elements Satisfy a Condition:** Find the indices in a 1D or 2D array where the elements satisfy a condition, such as being greater than a threshold.

   Input:

   ```
   arr = np.array([[10, 25, 17],
                   [22, 13, 30],
                   [5, 45, 20]])
   threshold = 20
   ```

   Expected Output:

   ```
   (array([0, 1, 1, 2]), array([1, 0, 2, 1]))
   ```

   Note: Also explain what the above output means.

4. **Random Sampling from a 1D Array:** Given a 1D array of size n, sample a set of m elements uniformly at random from the array.

   Input:

   ```
   arr = np.array([10, 20, 30, 40, 50, 60, 70, 80, 90])
   m = 4
   ```

   Expected Output: (Sample values will vary)

   ```
   [30, 10, 50, 80]
   ```

   Note: Now run your code for 10 times and check how the output behaves.

5. **Euclidean Distance Between Two Vectors:** Compute the Euclidean distance between two vectors.

   Input:

   ```
   a = np.array([1, 2, 3])
   b = np.array([4, 5, 6])
   ```

   Expected Output:

   ```
   5.196152422706632
   ```

6. **Broadcasting and Vectorization:** Add a scalar value to every element in a 2D array using broadcasting.

   Input:

   ```
   arr = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [7, 8, 9]])
   scalar = 5
   ```

   Expected Output:

```
[[ 6  7  8]
 [ 9 10 11]
 [12 13 14]]
```

7. **Solving a System of Equations:** Solve the system of linear equations given by:

$$2x + y = 5$$
$$x + 3y = 7$$

Input:

```
A = np.array([[2, 1],
              [1, 3]])
B = np.array([5, 7])
```

Expected Output:

```
[1. 2.]
```

Note: Make your code general so that any system of linear equations with m variables and n equations can be solved.

8. **Blood Pressure Analysis (Grouped Mean Calculation):** You have an array containing the blood pressure of a turtle every 6 hours (4 measurements per day). You've taken 100 measurements (25 days). Compute the mean blood pressure of the turtle for each day.

Input:

```
blood_pressure = np.random.randint(100, 200, size=100)
```

Expected Output: (Sample values will vary)

```
[150.5, 178.0, 160.75, ..., 142.5]
```

9. **Creating a 3D Array and Slicing:** Generate a 3D array of shape (3, 4, 5) filled with random integers from 0 to 50. Then, extract the second "layer" or "slice" of this array.

Input:

```
arr = np.random.randint(0, 50, size=(3, 4, 5))
```

Expected Output: (Sample values will vary)

```
array([[...], [...], [...], [...]])
```

Second Layer:

```
array([[...], [...], [...], [...]])
```

10. **Outer Addition and Multiplication (Broadcasting):** Given two 1D arrays, perform the following operations using broadcasting in NumPy:

    - Outer Addition: Add each element of array a to each element of array b and form a 2D matrix.
    - Outer Multiplication: Multiply each element of array a with each element of array b and form a 2D matrix.

    Input:

    ```
    a = np.array([1, 2, 3])
    b = np.array([4, 5, 6])
    ```

    Expected Output for Outer Addition:

    ```
    [[5 6 7]
     [6 7 8]
     [7 8 9]]
    ```

    Expected Output for Outer Multiplication:

    ```
    [[ 4  5  6]
     [ 8 10 12]
     [12 15 18]]
    ```

11. **Length of a Vector:** Compute the length (magnitude) of a vector.

    Input:

    ```
    v = np.array([3, 4, 5])
    ```

    Expected Output:

    ```
    7.0710678118654755
    ```

12. **Projection of a vector onto another:** Find the projection of vector **a** onto vector **b**.

    Formula:

    $$\mathbf{proj_b}(\mathbf{a}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2}\mathbf{b}$$

    Input:

    ```
    a = np.array([1, 2, 3])
    b = np.array([4, 5, 6])
    ```

    Expected Output: (Sample values will vary based on calculation)